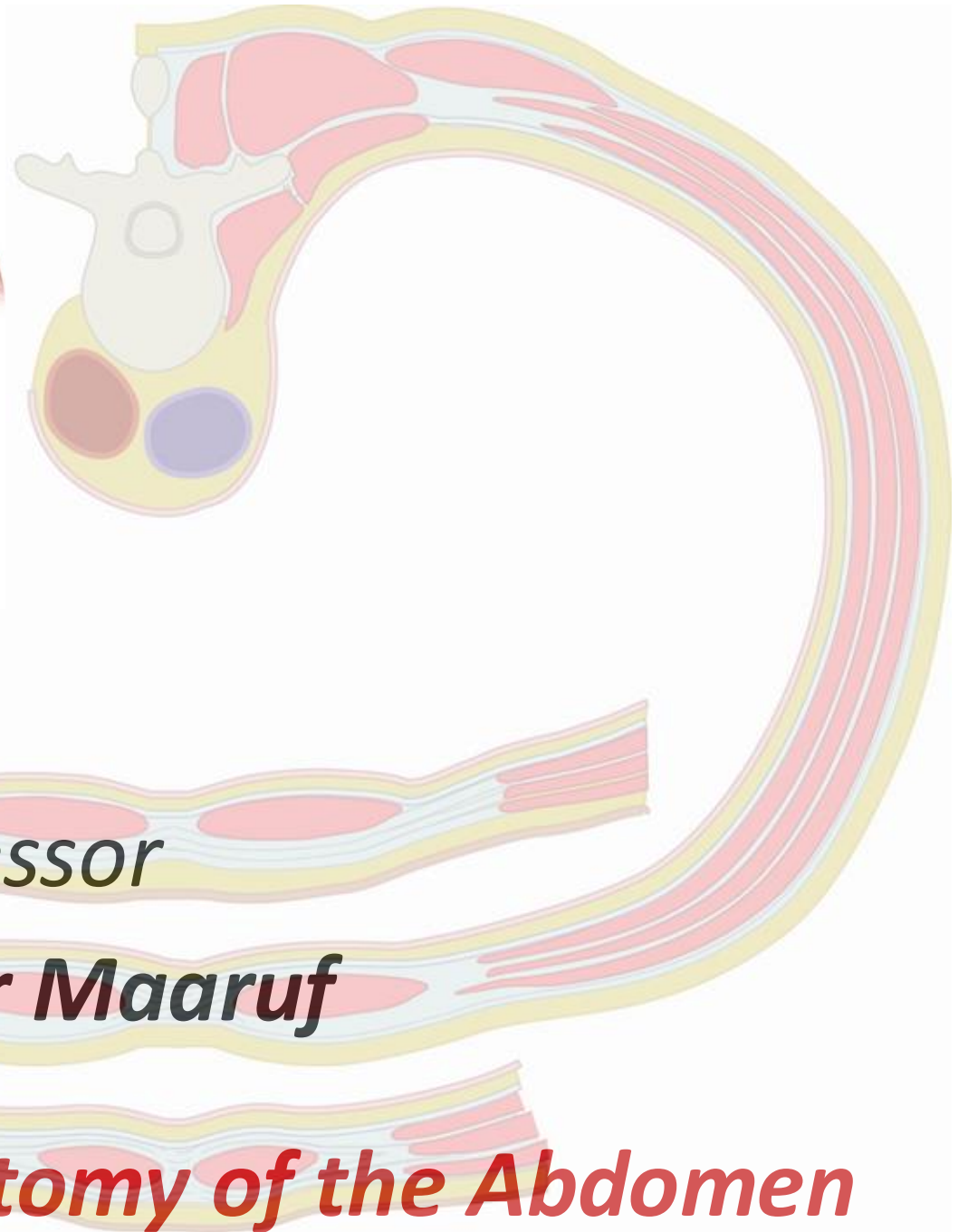
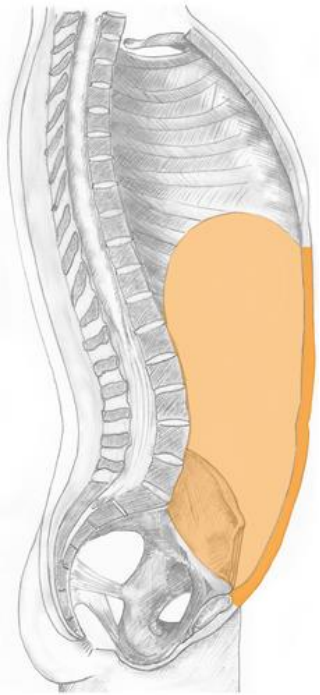


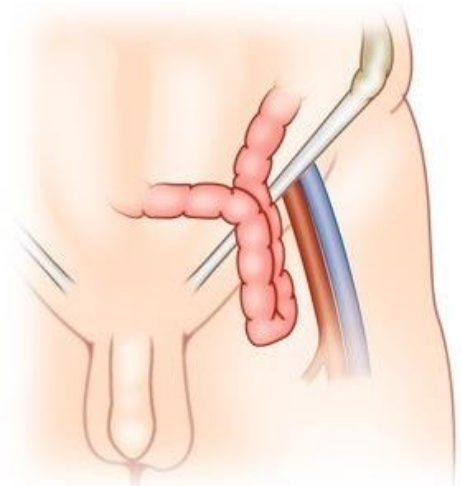


By:

Assistant Professor

Dr. Nali Abdulkader Maaruf

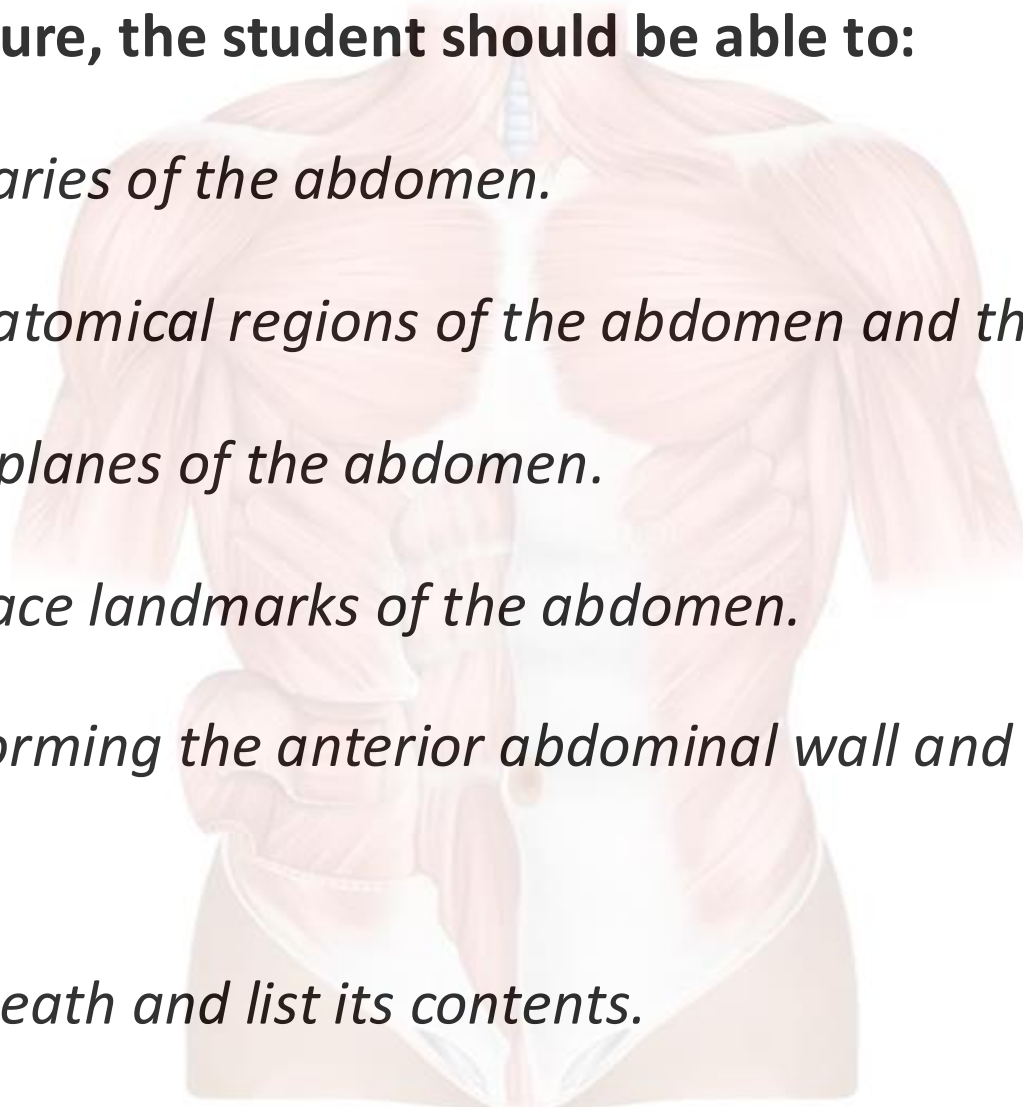
Anatomy of the Abdomen



Objectives

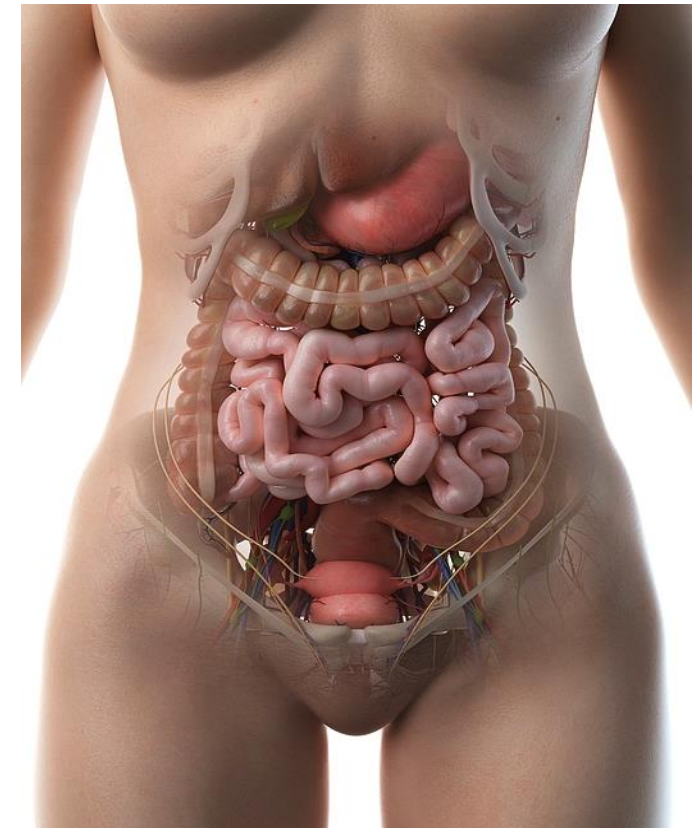
By the end of the lecture, the student should be able to:

- ☐ *Describe the boundaries of the abdomen.*
- ☐ *Identify the nine anatomical regions of the abdomen and their contents.*
- ☐ *List the anatomical planes of the abdomen.*
- ☐ *Enumerate the surface landmarks of the abdomen.*
- ☐ *Outline the layers forming the anterior abdominal wall and explain the clinical correlations of each.*
- ☐ *Define the rectus sheath and list its contents.*



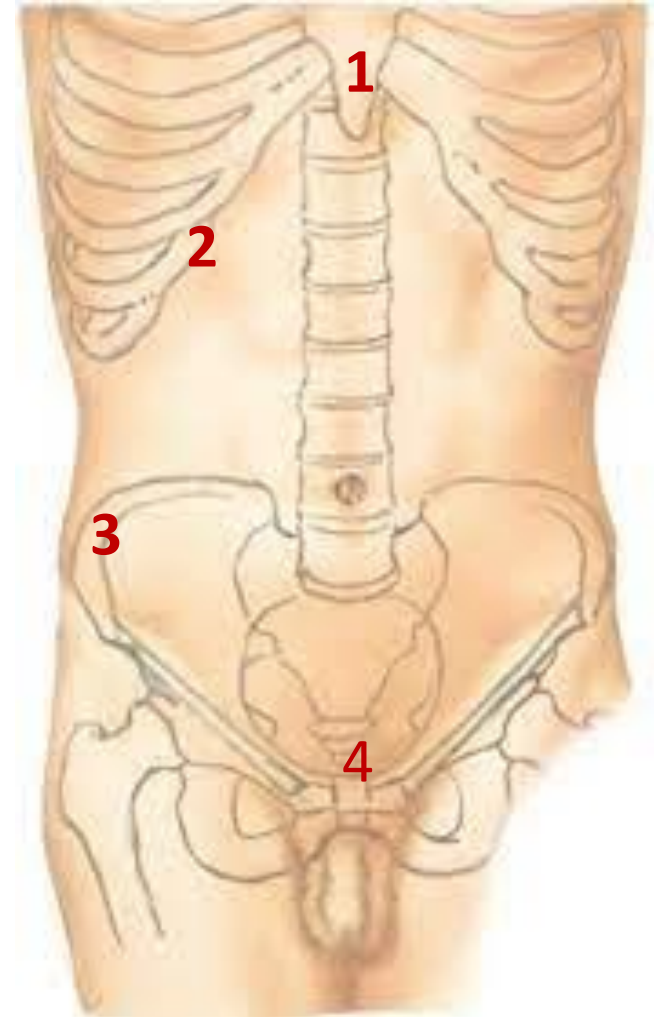
ABDOMEN

- ❑ The abdomen is the lower part of the trunk located below the diaphragm. Its walls enclose a large cavity of the body known as the abdominal cavity.
- ❑ Clinically, the abdomen is of significant importance, and a physical examination of a patient cannot be considered complete without its assessment.
- ❑ The boundaries of the abdominal cavity are:
 - **Superiorly:** is a diaphragm, which separates it from the thoracic cavity.
 - **Inferiorly:** It continues with the pelvic cavity at the pelvic inlet.
 - **Anteriorly:** Is the anterior abdominal wall.
 - **Posteriorly:** Posterior abdominal wall



Surface landmarks of abdomen

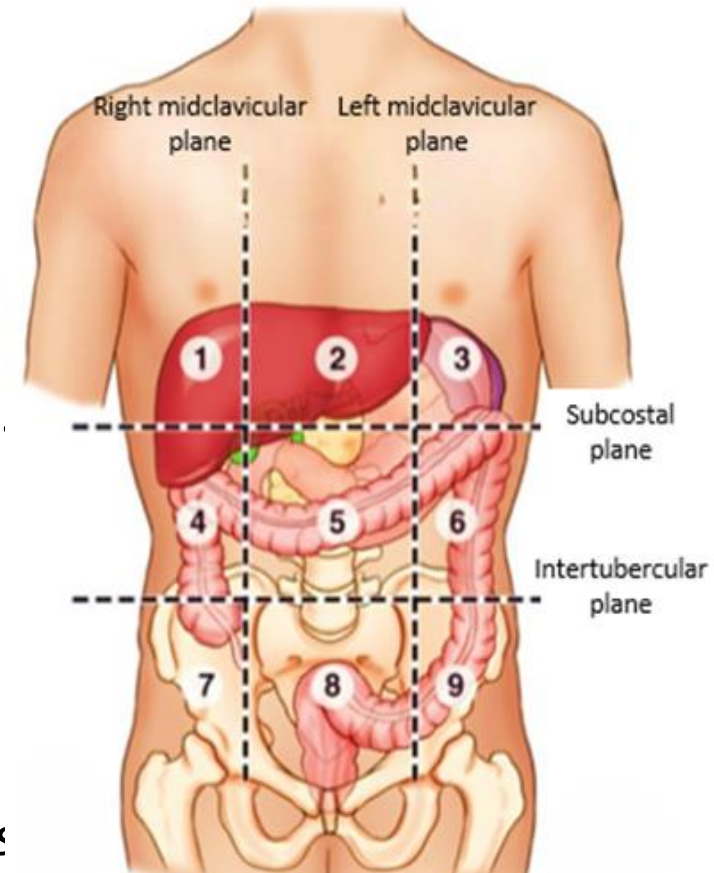
1. **Costal Margins:** The costal margins are formed by the cartilages of the 7th, 8th, 9th, and 10th ribs.
2. **Xiphoid Process:** It is located at the apex of the subcostal angle and corresponds to the level of the T9 vertebra.
3. **Iliac Crest:** The highest points of the iliac crests are situated at the level of the L4 vertebra, just below the umbilicus, which delineates a plane running through the L3-4 (Inter-crystal plane).
4. **Pubic Symphysis:** This landmark represents the lower limit of the anterior abdominal wall in the anterior median plane. It is easily palpable and is located at the level of the coccyx.



Nine Regions of Abdomen

- ❑ The abdominal cavity is divided into nine regions by two horizontal and two vertical planes
- ❑ The superior horizontal plane (the subcostal plane) passes from the lower border of the costal cartilage of the 10th rib anteriorly and vertebra L3 posteriorly.
- ❑ The inferior horizontal plane (the intertubercular plane) extends from the tubercles of the iliac crests at the level of L5.
- ❑ The vertical planes run from the midpoint of the clavicles inferiorly to a point midway between the anterior superior iliac spine and the symphysis.

The abdominal cavity houses the majority of the digestive system organs, as well as portions of the urinary system and parts of the endocrine glands. These structures are organized according to the nine abdominopelvic regions



Contents Nine Regions of Abdomen

<u>1) Right hypochondrium</u> • Right lobe of liver • Gall bladder • Right colic flexure • Right suprarenal gland • Upper part of right kidney	<u>2) Epigastrium</u> • Pyloric part of stomach • Liver • Pancreas • Duodenum • Abdominal aorta	<u>3) Left hypochondrium</u> • Spleen, • Left kidney, • Left suprarenal • Stomach, • Pancreas, • Splenic flexure of colon.
<u>4) Right Lumbar</u> • gallbladder, • Right kidney, • Part of the liver, and Ascending colon	<u>5) Umbilicus</u> • Omentum • Mesentery • Transverse colon • Duodenum, jejunum, and the ileum	<u>6) Left lumbar</u> • Descending colon, the left kidney, • Parts of the jejunum and ilium • Part of the spleen
<u>7) Right iliac</u> • Cecum • Appendix • Right ureter and Right ovary	<u>8) Hypogastric or Pubic</u> • Ilium • Bladder	<u>9) Left iliac</u> • Descending colon, • Sigmoid colon • Left ureter and Left ovary

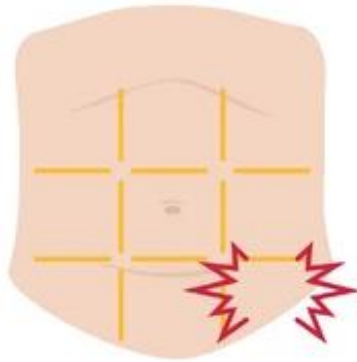


CLINICAL

CASE

A 60-year-old man presents with left lower abdominal pain (left iliac fossa) associated with constipation and mucus in stools. Colonoscopy shows diverticulosis. Which organ is most likely the source of the pain?

- A. Pancreas
- B. Sigmoid colon
- C. Appendix
- D. Small intestine
- E. Gallbladder

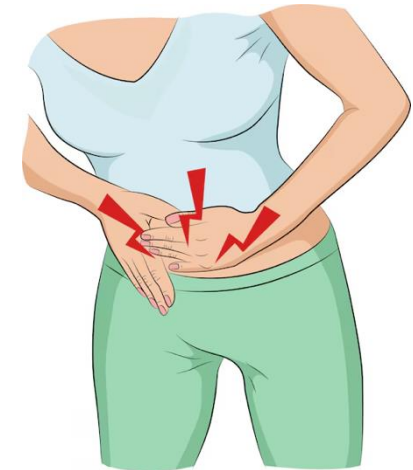


SCENARIO



A 30-year-old female presents with colicky pain in the umbilical region, associated with abdominal distension and altered bowel habits. Which structure is most likely involved?

- A. Small intestine
- B. Sigmoid colon
- C. Right kidney
- D. Gallbladder
- E. Appendix



ABDOMINAL PLANES

The abdominal planes are crucial imaginary horizontal and vertical lines used in anatomy and clinical practice for descriptive, diagnostic, or surgical purposes, as well as for describing pain and tenderness.

❑ Trans umbilical plane

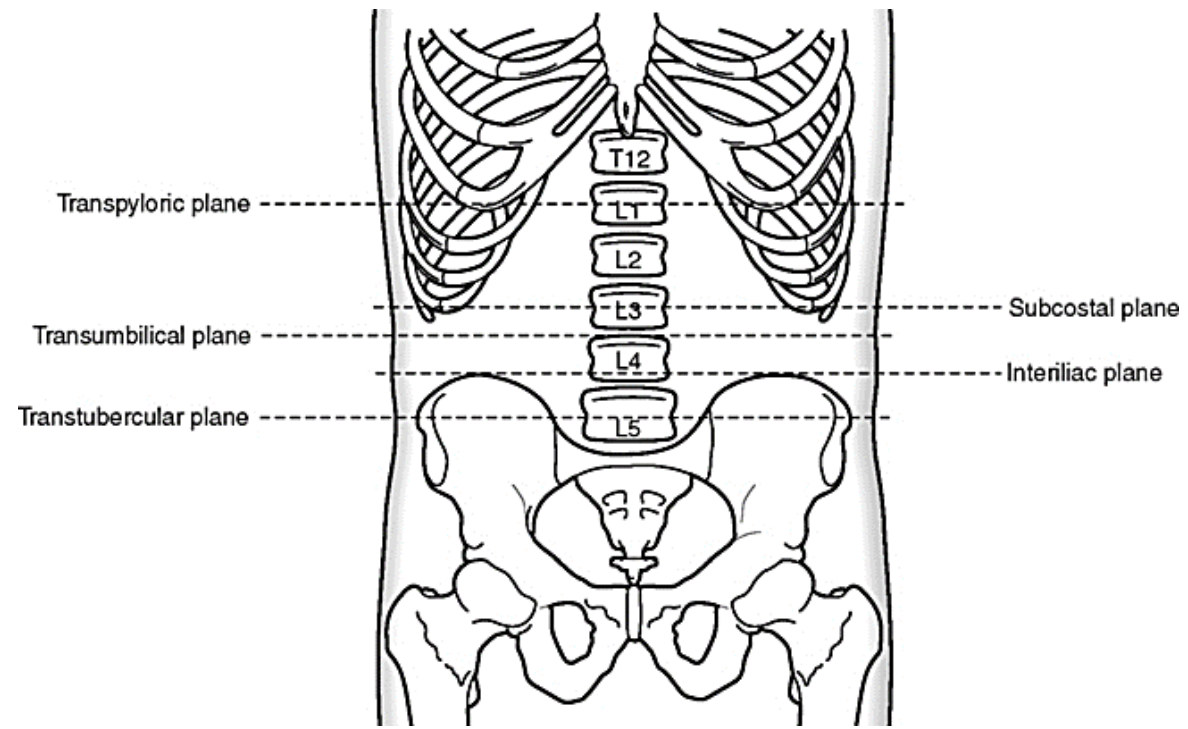
This transverse plane passes through the umbilicus (navel) and is located at the level of the intervertebral disc between the L3 and L4 vertebrae.

❑ Subcostal Plane

❑ Intertubercular Plane

❑ Right and Left Vertical Planes (Midclavicular Planes)

❑ Transpyloric Plane/Addison's Plane

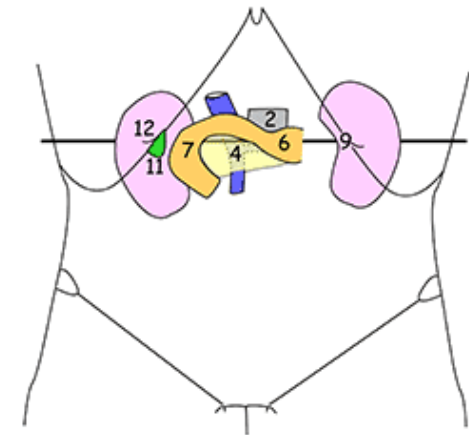
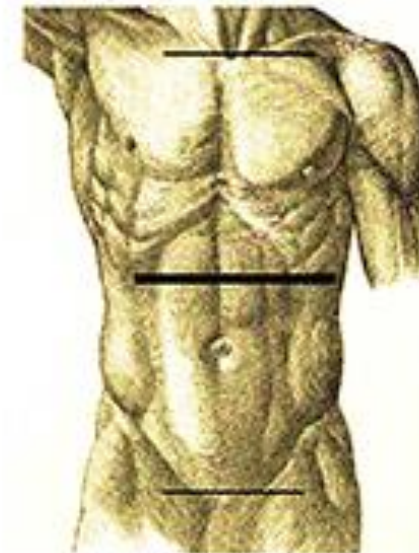


The transpyloric plane: is an imaginary horizontal plane that extends from the midpoint between the jugular (suprasternal) notch and the symphysis pubis.

Anteriorly, this plane passes through the tip of the 9th costal cartilage, while posteriorly, it intersects the lower border of the body of the L1 vertebra.

This plane is significant in abdominal anatomy, as it corresponds to several abdominal viscera:

1. Pylorus in the stomach
2. Neck of the pancreas
3. Duodeno-jejunal flexure
4. Fundus of the gallbladder
5. Hilum of the kidneys
6. Origins of the (superior mesenteric artery and portal vein)
7. Tip of the ninth costal cartilage
8. Termination of the spinal cord (conus medullaris)

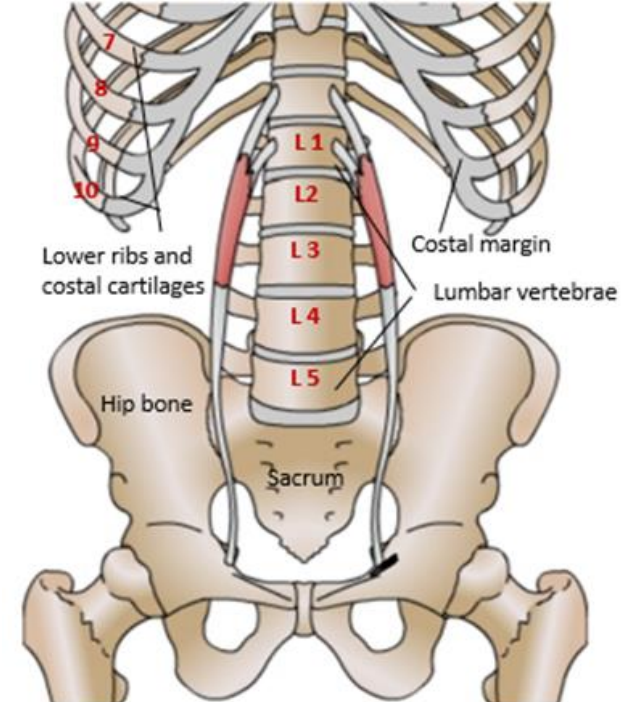


ABDOMINAL WALLS



The dynamic shield and functional corset of the abdomen
— Claudius Galen

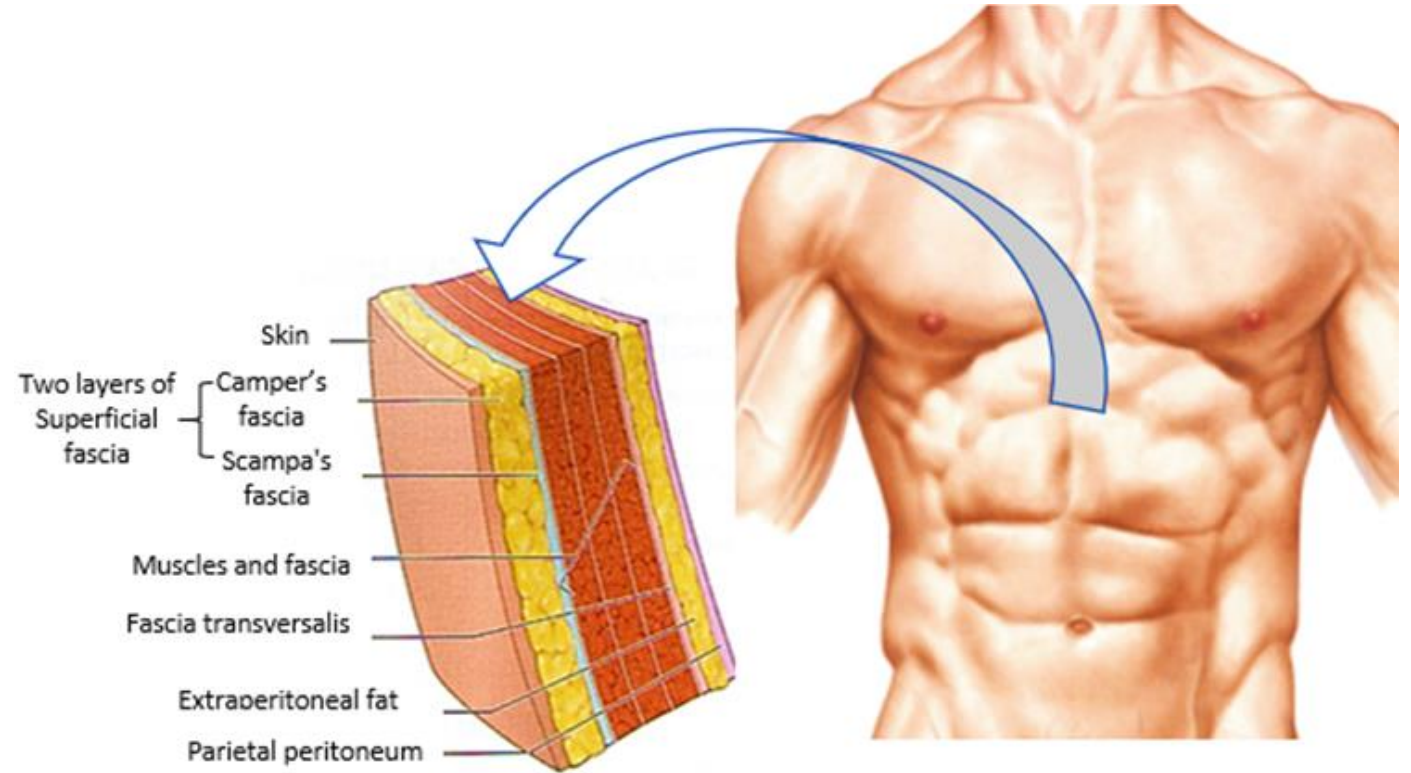
- ❑ Anterior and posterior abdominal walls separate from each other by the midaxillary line.
- ❑ The anatomy of the anterior abdominal wall is clinically significant because:
 - The physical examination of the abdomen is primarily conducted through the anterior abdominal wall.
 - Access to the abdomen and its contents is typically achieved via incisions made in the anterior abdominal wall.
 - Abdominal hernias predominantly occur through this wall.
- ❑ The following bones constitute the abdomen and pelvis
 - Lower ribs and costal cartilages.
 - Lumbar vertebrae, Sacrum and Coccyx.
 - Hip bone (also known as the innominate bone).



Anterior abdominal wall

The anterior abdominal wall comprises six distinct layers, from superficial to deep are:

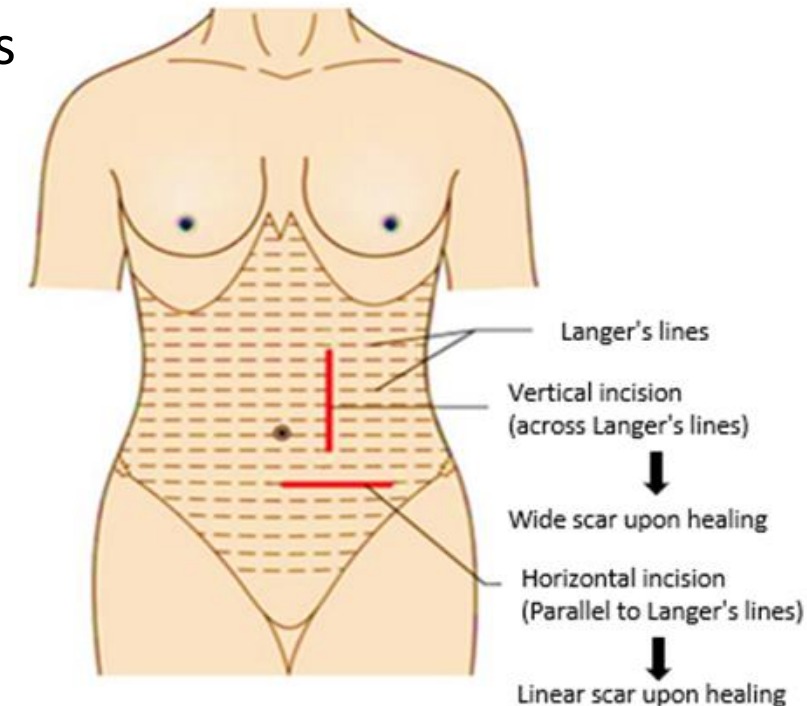
- 1) Skin
- 2) Superficial fascia
- 3) Muscles
- 4) Fascia transversalis
- 5) Extraperitoneal tissue
- 6) Parietal layer of peritoneum



- Notably, the deep fascia is absent in the anterior abdominal wall, which allows for the bulging or distension of the abdominal wall after meals, during pregnancy, and in other circumstances. This absence is also observed in the penis, scrotum, face and perineum.

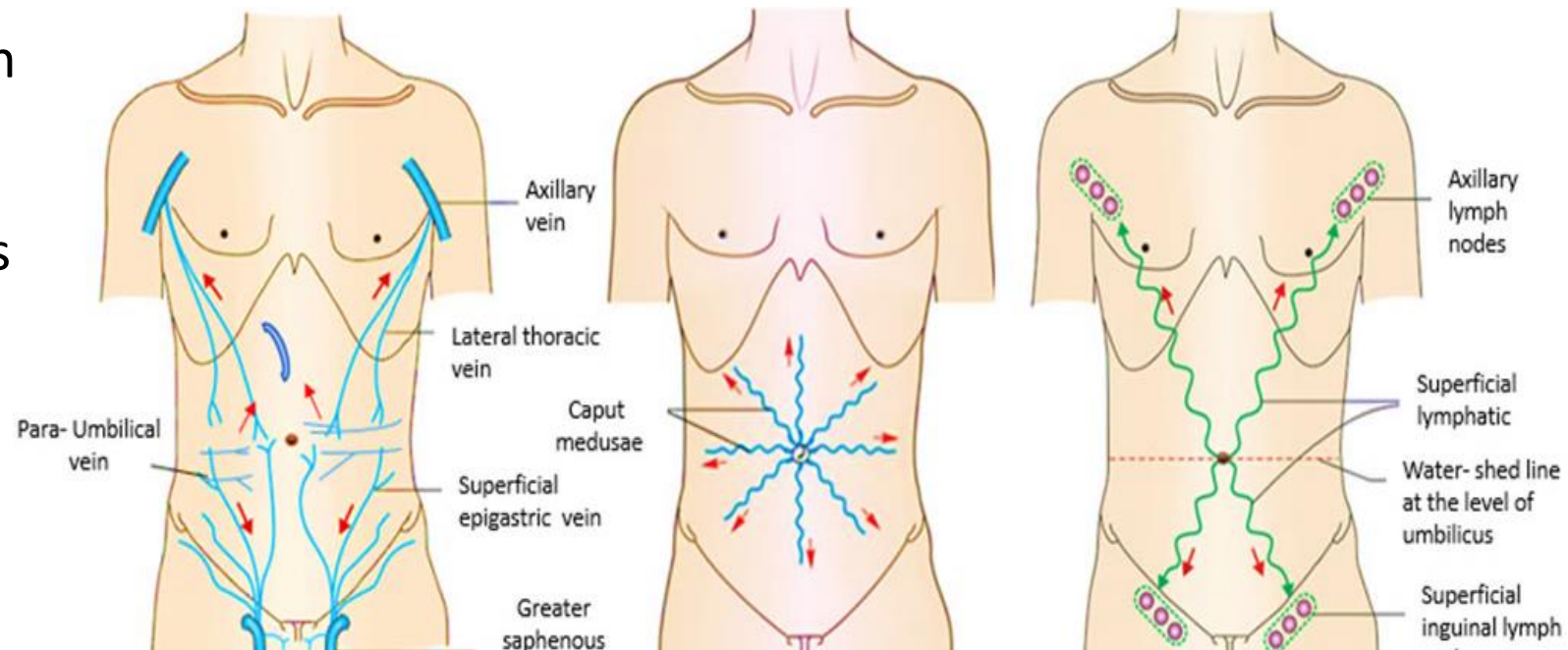
1. SKIN

- ❑ The skin is loosely attached to the underlying structures, except at the umbilicus
- ❑ The cleavage lines (Langer's lines) in the anterior abdominal wall run horizontally
- ❑ The umbilicus is of significant anatomical, embryological, and clinical importance, as it is the most prominent feature of the anterior abdominal wall.
- ❑ It represents the site of attachment for the umbilical cord in the fetus
- ❑ The position of the umbilicus is variable:
 - In adults, it lies at the level L3 and L4 vertebrae.
 - In newborns, it is slightly lower due to a less developed pelvic region.
 - In old age, it descends to a lower-level owing to decreased tone of the abdominal muscles.



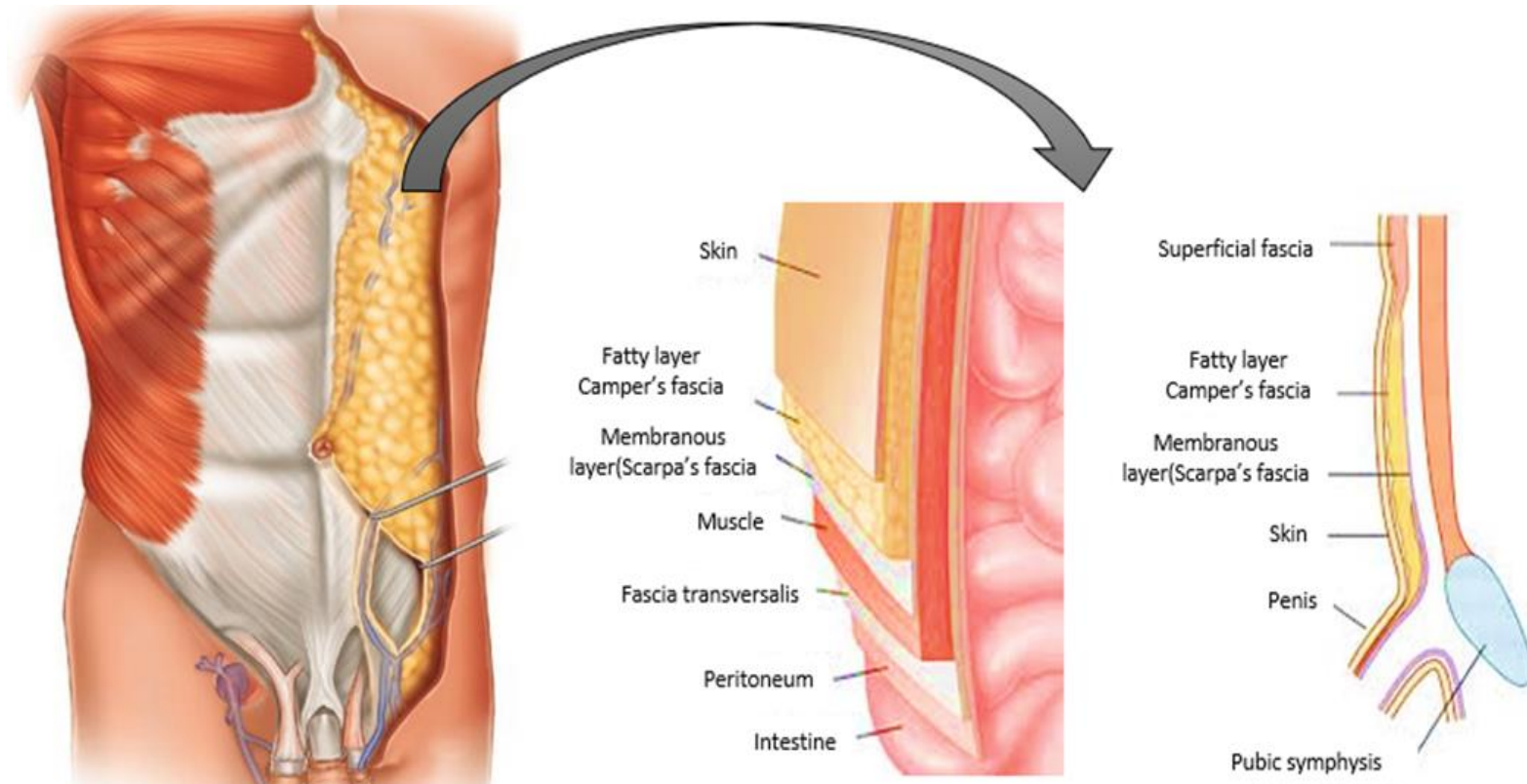
Anatomical significance of the umbilicus

- ❑ The level of the umbilicus serves as a watershed line for arterial, venous, and lymphatic drainage
- ❑ Venous drainage above the umbilicus passes into the axillary vein via the lateral thoracic vein, below it enters the femoral vein via the superficial epigastric and great saphenous veins.
- ❑ Lymphatics above the umbilicus drain into the axillary lymph nodes, below drain into the superficial inguinal nodes
- ❑ The umbilicus indicates the level of the T10 dermatome
- ❑ It is one of the important site of portocaval anastomosis
- ❑ Although the umbilicus is a scar on the anterior abdominal wall, it holds significant cosmetic value, especially in females. Consequently, surgical incisions should never be made across it.



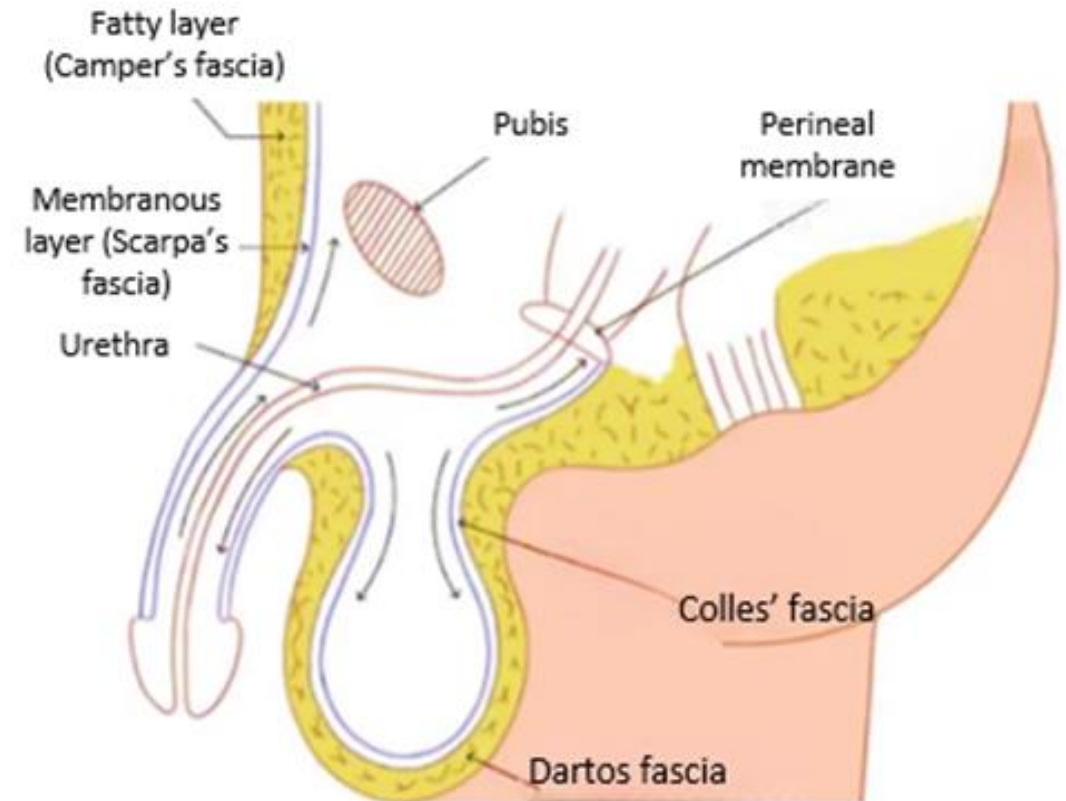
2. SUPERFICIAL FASCIA

- ❑ The superficial fascia of the abdominal wall is a layer of fatty connective tissue, typically existing as a single layer that is similar to, and continuous with, the superficial fascia of other regions of the body.
- ❑ However, below the umbilicus the superficial fascia divides into two layers: a superficial fatty layer (*Camper's fascia*) and a deeper membranous layer (*Scarpa's fascia*).
- ❑ The space between the two layers contains the superficial epigastric, circumflex iliac, and external pudendal vessels, as well as superficial inguinal lymph nodes



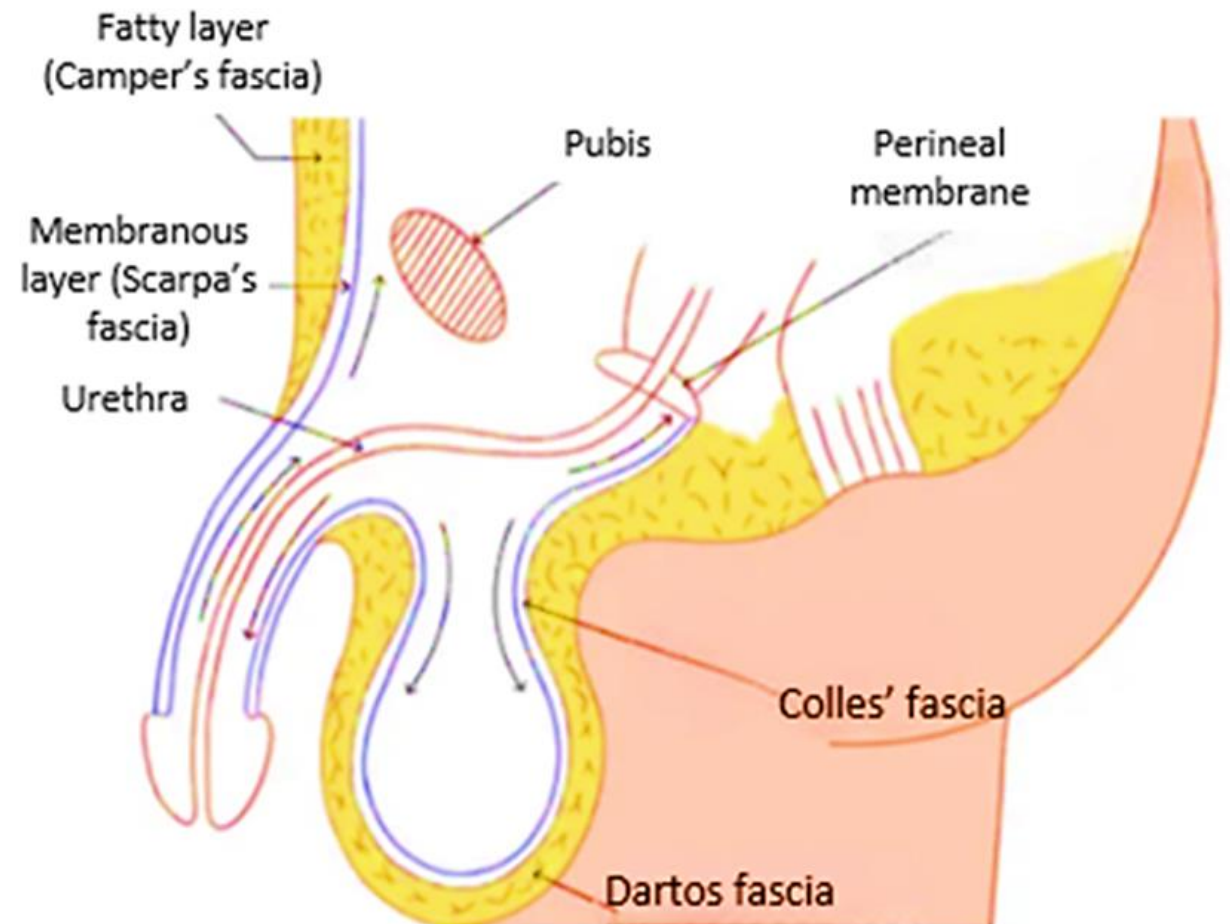
Camper's Fascia (Superficial Fatty Layer)

- ❑ The superficial fatty layer of the superficial fascia, known as Camper's fascia, contains fat and varies in thickness
- ❑ It is continuous over the inguinal ligament with the superficial fascia of the thigh and with a similar layer in the perineum.
- ❑ In men, this superficial layer extends over the penis and, as it loses fat, continues into the scrotum, where it forms a specialized fascial layer containing smooth muscle called the Dartos fascia.
- ❑ In women, this superficial layer retains some fat and is a component of the labia majora.



Scarpa's fascia

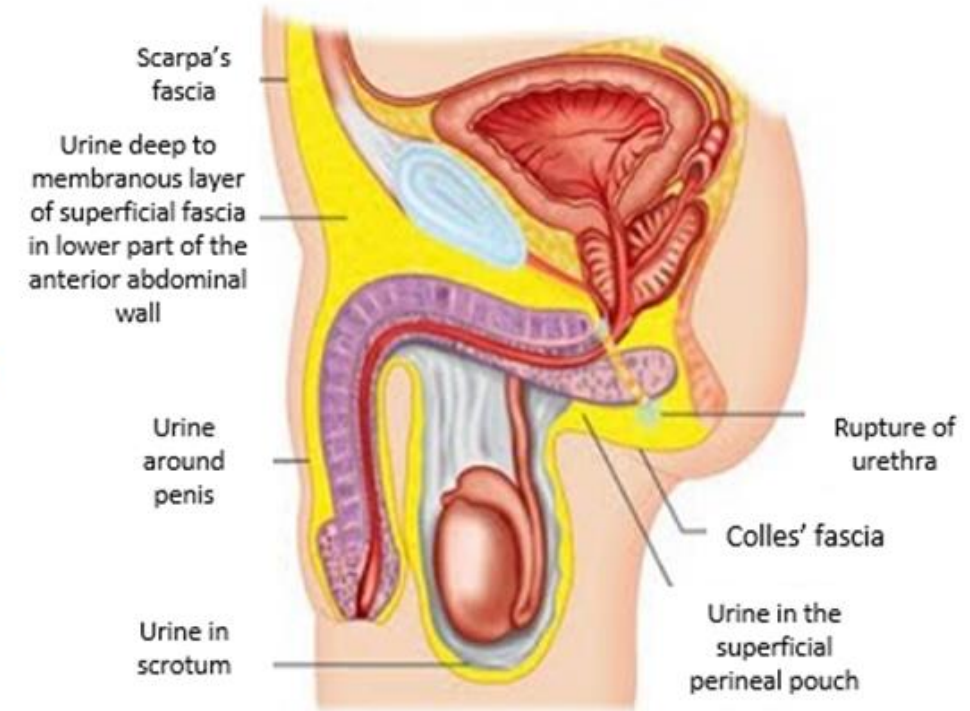
- ❑ The deeper membranous layer of superficial fascia, known as Scarpa's fascia, is thin and contains little to no fat.
- ❑ It is prolonged over the penis, enclosing it and forming the fascia of the penis (Buck's fascia).
- ❑ It covers the scrotum and continues with the superficial fascia of the perineum, where it is called Colle's fascia.



CLINICAL CORRELATION

The membranous layer of the superficial fascia is clinically significant because beneath it lies a potential closed space that does not open into the thigh but connects with the superficial perineal pouch via the penis and scrotum.

Rupture of the penile urethra may lead to extravasation of urine into the Scrotum, Perineum, Penis, and Lower part of the anterior abdominal wall, deep to the membranous layer of fascia.



The urine does not enter the thigh due to the attachment of the fascia to the fascia lata.



CLINICAL

CASE

SCENARIO



A 50-year-old obese male presents with a swelling in the lower anterior abdominal wall, diagnosed as a hernia. During surgery, the surgeon identifies the superficial fascia composed of two layers. The outer fatty layer shows variable thickness depending on the patient's nutritional status. Which layer is this?

- A. Camper's fascia
- B. Scarpa's fascia
- C. Transversalis fascia
- D. External oblique aponeurosis
- E. Rectus sheath

A 35-year-old man presents with scrotal swelling. On examination, the surgeon recalls that the superficial fascia of the lower anterior abdominal wall continues into the penis and scrotum. In the scrotum, this layer loses its fat and forms a specialized fascia containing smooth muscle fibers. Which fascia is being described?

- A. Camper's fascia
- B. Scarpa's fascia
- C. Dartos fascia
- D. Buck's fascia
- E. Colles' fascia

3.MUSCLES

The muscles of the abdominal wall are divided into two groups:

A) Flat muscles: These 1. External oblique, 2. Internal oblique and 3. Transversus abdominis

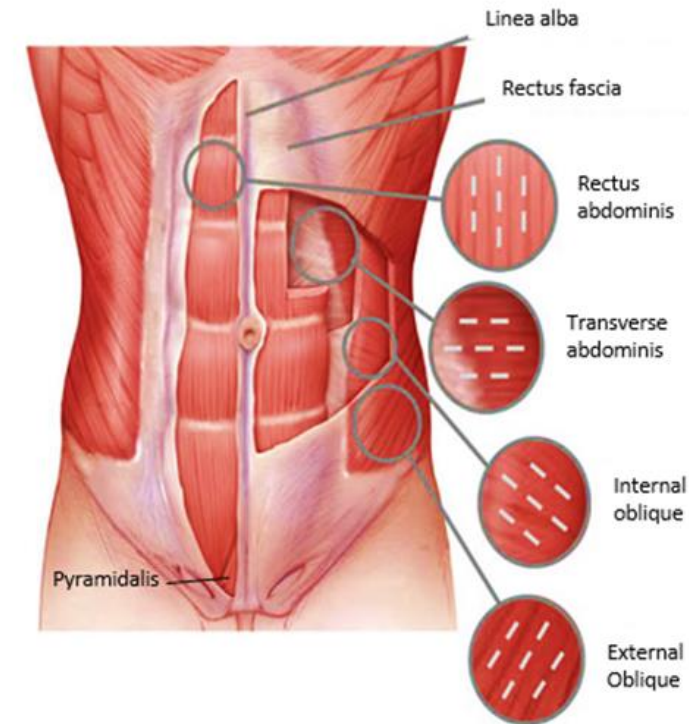
They aponeurotic anteriorly to form the rectus sheath.

B) Vertical muscles: These include the bilateral

1. Rectus abdominis and 2. Pyramidalis

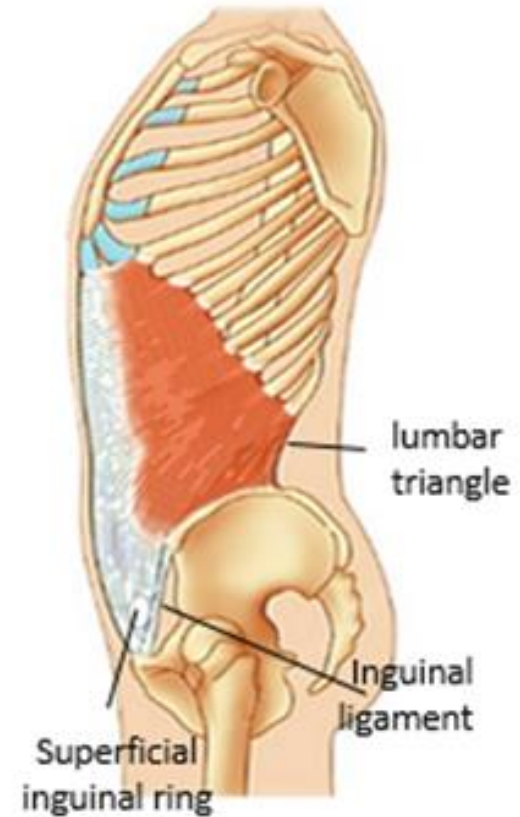
The three main functions of the anterior abdominal muscles are:

- To provide strong and expandable support for the abdominal viscera
- To increase intra-abdominal pressure, thereby assisting in defecation, micturition and delivery
- To facilitate trunk movement to maintain posture.



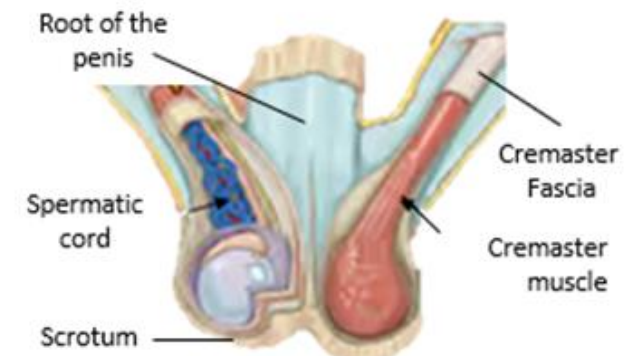
External Oblique Muscle

- ❑ This is the most superficial muscle, with most fibers running downward, forward, and medially, similar to the motion of placing a hand inside the side pocket of pants.
- ❑ The free lower border of the aponeurosis is thickened and rolled inward on itself, forming the inguinal ligament.
- ❑ Above the pubic symphysis, the aponeurosis presents a triangular gap known as the superficial inguinal ring.
- ❑ The posterior free border of the muscle forms the anterior boundary of the lumbar triangle.



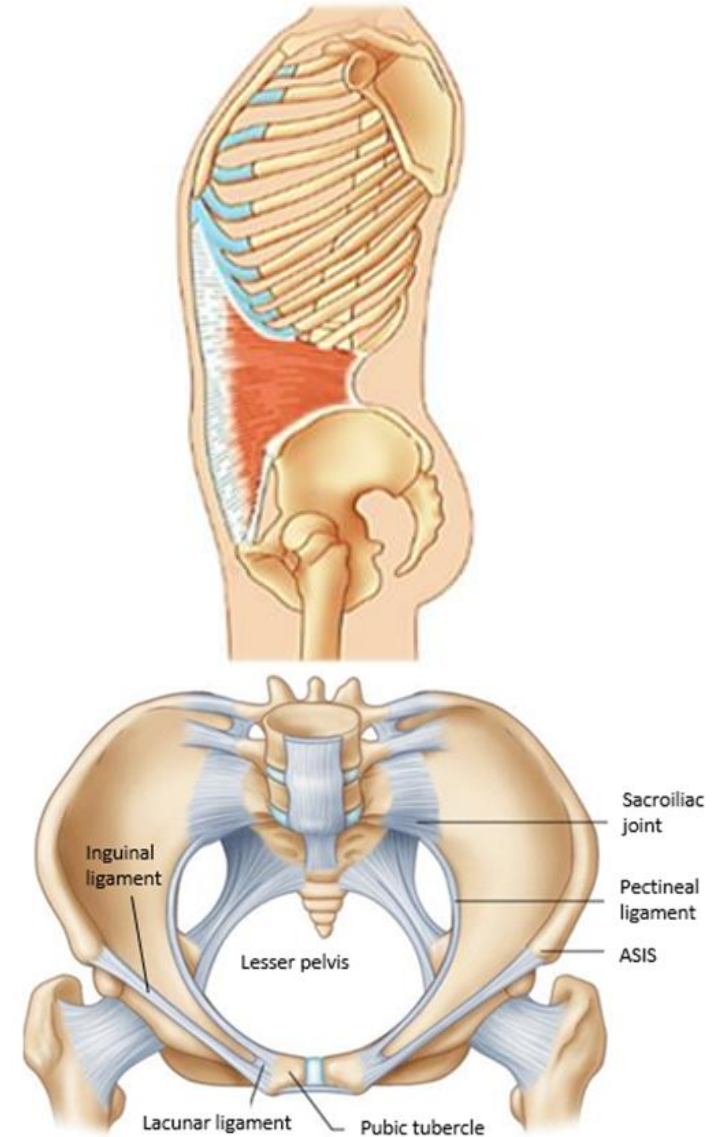
Internal Oblique Muscle

- ❑ This muscle is relatively smaller and thinner than the external oblique.
- ❑ It lies deep to the external oblique,
- ❑ Most fibers run upward, forward, and medially, akin to the motion of placing a hand in front of the chest.
- ❑ The fibers arise from the inguinal ligament, arch over the inguinal canal, and fuse with the corresponding tendinous fibers of the transversus abdominis muscle to form the conjoint tendon, which is inserted into the pubic crest and the medial part of the pecten pubis
- ❖ The cremaster muscle and fascia derive from the lower arched fibers of the internal oblique, covering the spermatic cord and testis.



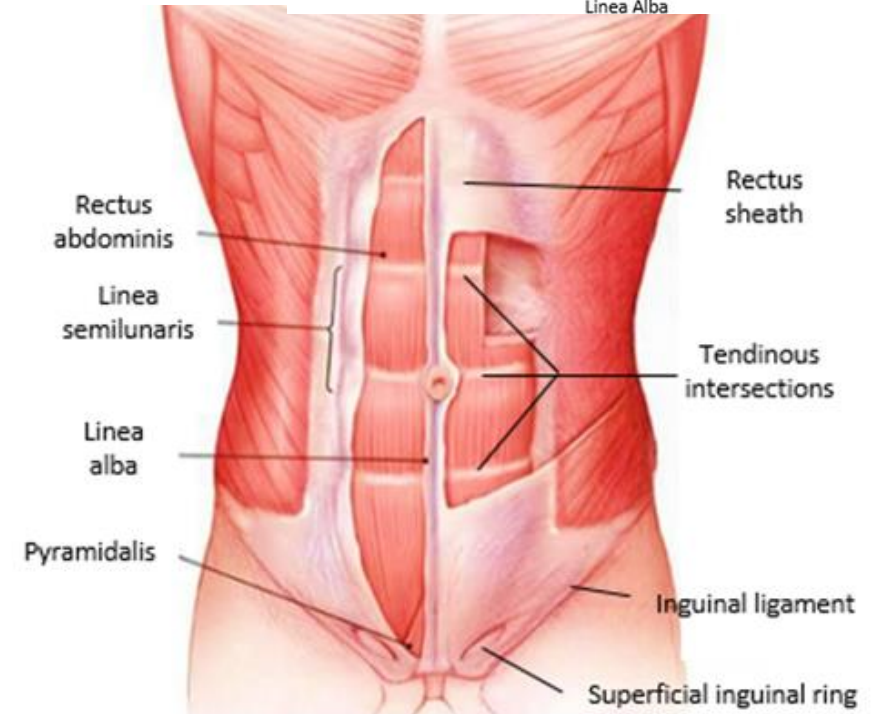
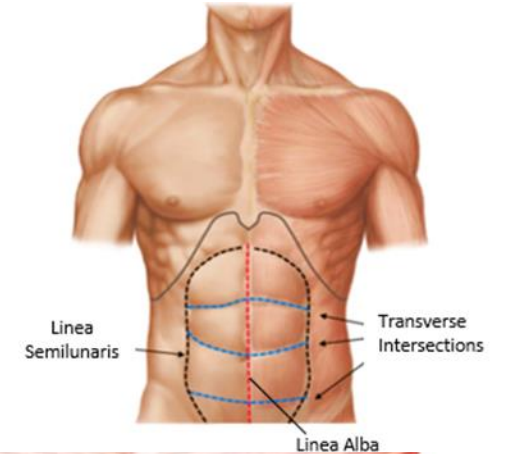
Transversus Abdominis Muscle

- ❑ The transversus abdominis muscle is the innermost of the three flat abdominal muscles, with most of its fibers oriented horizontally or transversely.
- ❑ Structures derived from flat muscles
- ❑ The structures derived from flat muscles are as follows:
 - Inguinal ligament** is a thick, fibrous band that extends from the anterior superior iliac spine to the pubic tubercle.
 - Conjoint tendon.**
 - Cremaster muscle**



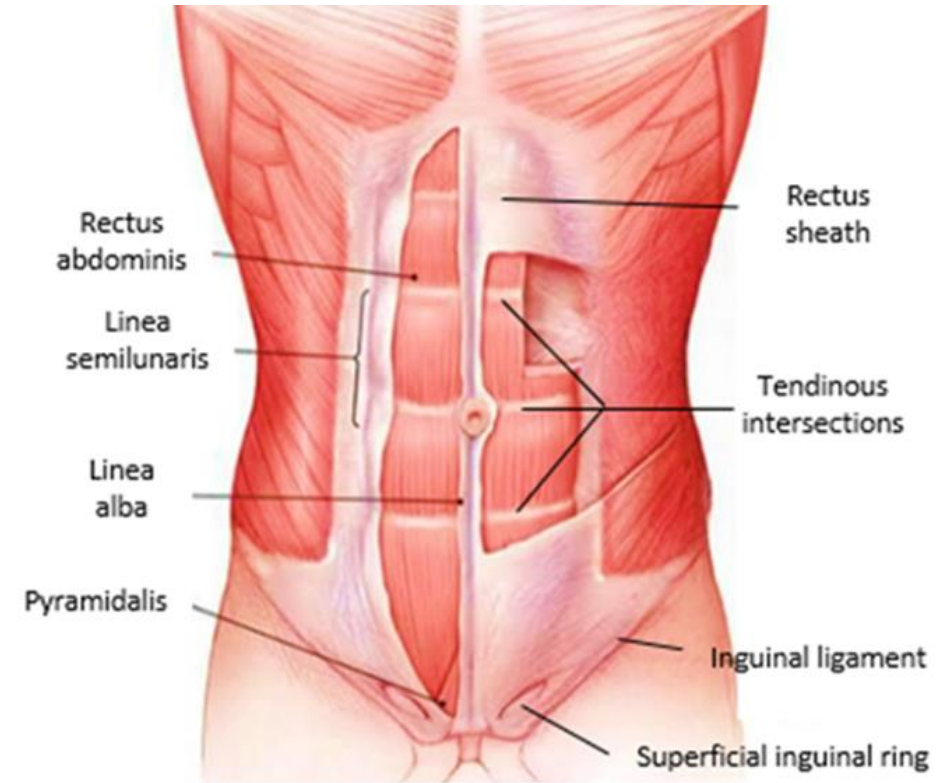
Rectus Abdominis Muscle

- ❑ This is a long muscle that extends vertically upward along the Linea alba, from the pubic symphysis below to the costal margin above.
- ❑ The lateral border of the rectus abdominis is referred to as the Linea semilunaris.
- ❑ The muscle features three tendinous intersections:
 - The first is at the level of the tip of the xiphoid process.
 - The second is at the level of the umbilicus.
 - The third is located midway between the first two.
 - Occasionally, a fourth may be present below the umbilicus.



Pyramidalis

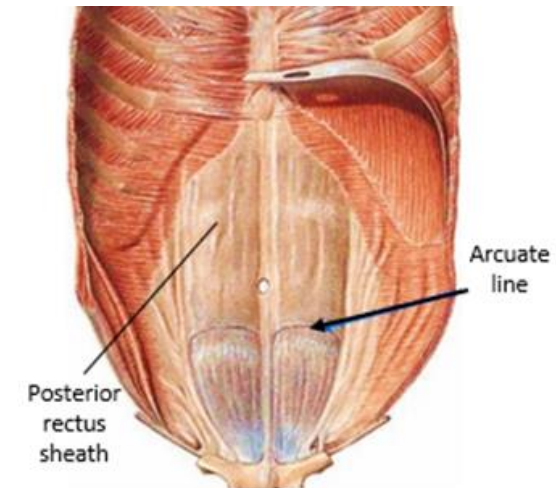
- The pyramidalis is considered rudimentary in humans.
- It is a small triangular muscle located anterior to the lower part of the rectus abdominis muscle within the rectus sheath, and it is absent in approximately 20% of individuals.



Muscle	Origin	Insertion	Innervation	Main Action
External oblique	External surfaces of 5th–12th ribs	Linea alba, pubic tubercle, and iliac crest	Thoraco-abdominal nerves (T7 to T12)	Flexes and rotates trunk
Internal oblique	Thoracolumbar fascia and lateral third of inguinal ligament	10th–12th ribs, linea alba, and pecten pubis via conjoint tendon	Thoraco-abdominal and first lumbar nerves	Flexes and rotates trunk
Transversus abdominis	Thoracolumbar fascia, iliac crest, lateral third of inguinal ligament	Linea alba, and pecten pubis via conjoint tendon	Thoraco-abdominal and first lumbar nerves	Supports abdominal viscera
Rectus abdominis	Pubic symphysis and pubic crest	Xiphoid process and 5th–7th costal cartilages	Thoraco-abdominal nerves	Flexes trunk

RECTUS SHEATH

- ❑ The rectus sheath is an aponeurotic sheath that encloses two muscles, two nerves, and two blood vessels. It is derived from the aponeuroses of the flat muscles of the anterior abdominal wall
- ❑ The inferior margin of the posterior leaflet of the rectus sheath within the abdomen is known as the arcuate line
- ❑ This line is located midway between the umbilicus and the pubic symphysis and marks the point where the posterior rectus sheath ends.
- ❑ Below this line, a structurally weaker region is created, which plays a role in the distribution of abdominal pressure and explains the increased susceptibility of the lower abdomen to hernias.





CLINICAL

CASE

SCENARIO

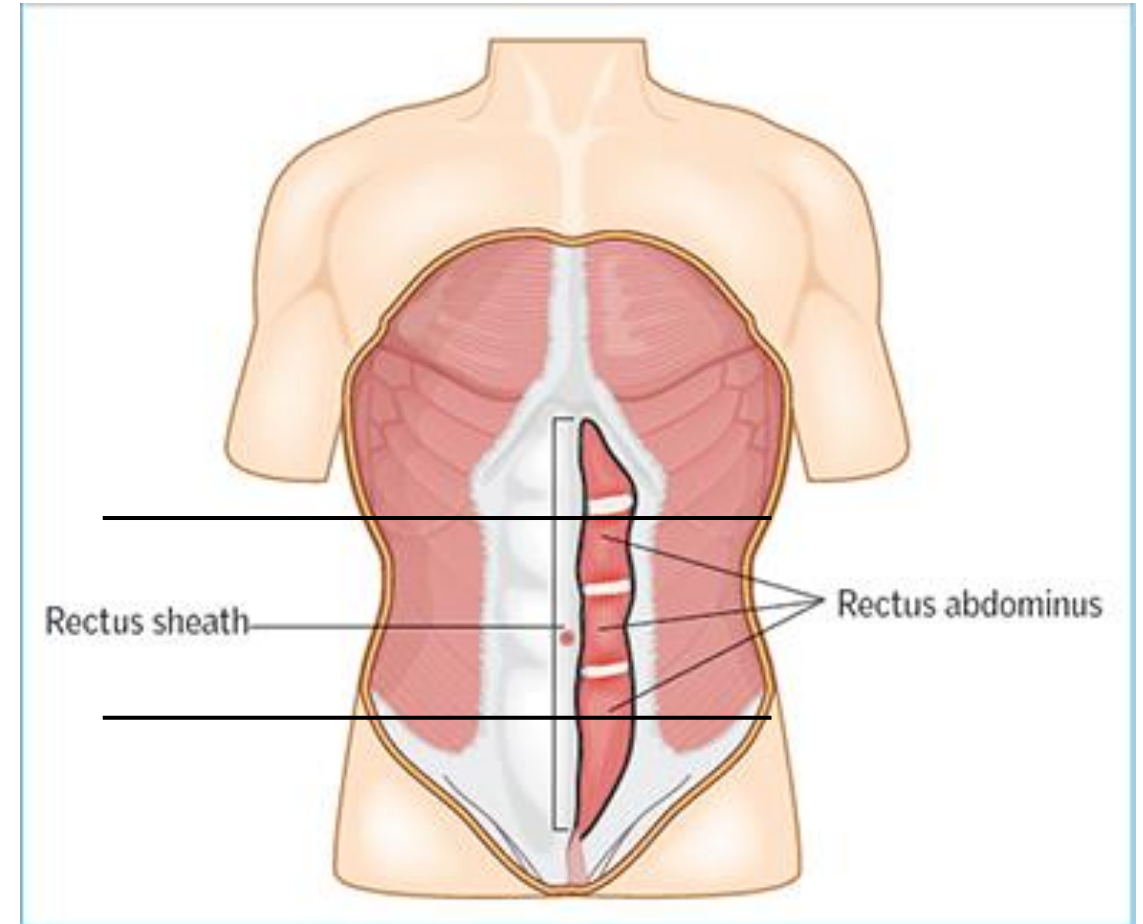


During hernia repair, the surgeon notes the posterior rectus sheath ends at a curved margin, leaving the area structurally weaker and more prone to hernia. where is the anatomical location of this area

- A. Around Linea alba
- B. Lateral to Linea semilunaris
- C. Below Arcuate line
- D. Medial to the inguinal ligament
- E. Above the transpyloric line

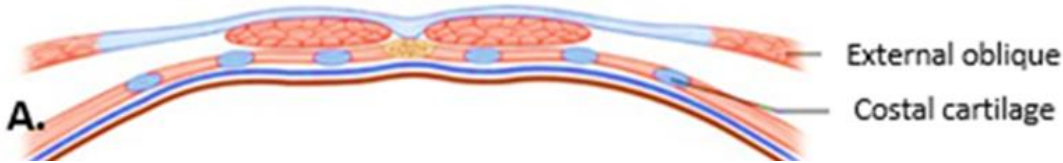
- For anatomical and surgical purposes, the rectus sheath is typically divided into three levels based on the arrangement of the aponeurotic layers surrounding the rectus abdominis muscle.

- A. Above the level of the costal margin:
- B. Between the costal margin and the arcuate line:
- C. Below the level of the arcuate line:



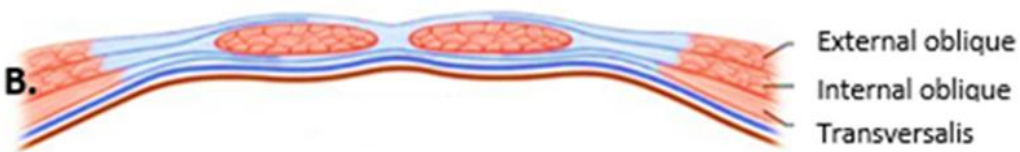
A. Above the level of the costal margin:

- Anterior wall is formed by the aponeurosis of the external oblique
- Posterior wall is formed by the 5th, 6th, and 7th costal cartilages.



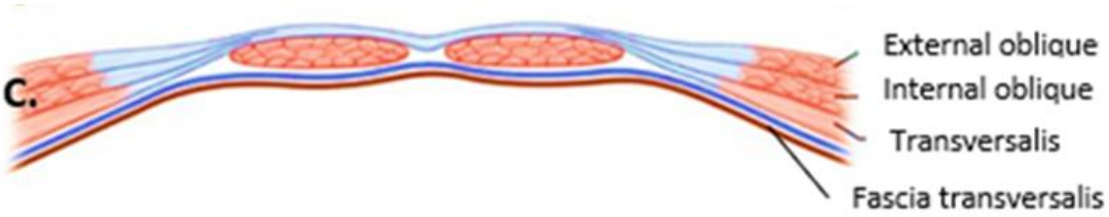
B. Between the costal margin and the arcuate line:

- Anterior wall is formed by the fusion of the aponeurosis of the external oblique with the anterior lamina of the aponeurosis of the internal oblique.
- Posterior wall is created by the fusion of the aponeurosis of the transversus abdominis with the posterior lamina of the aponeurosis of the internal oblique.



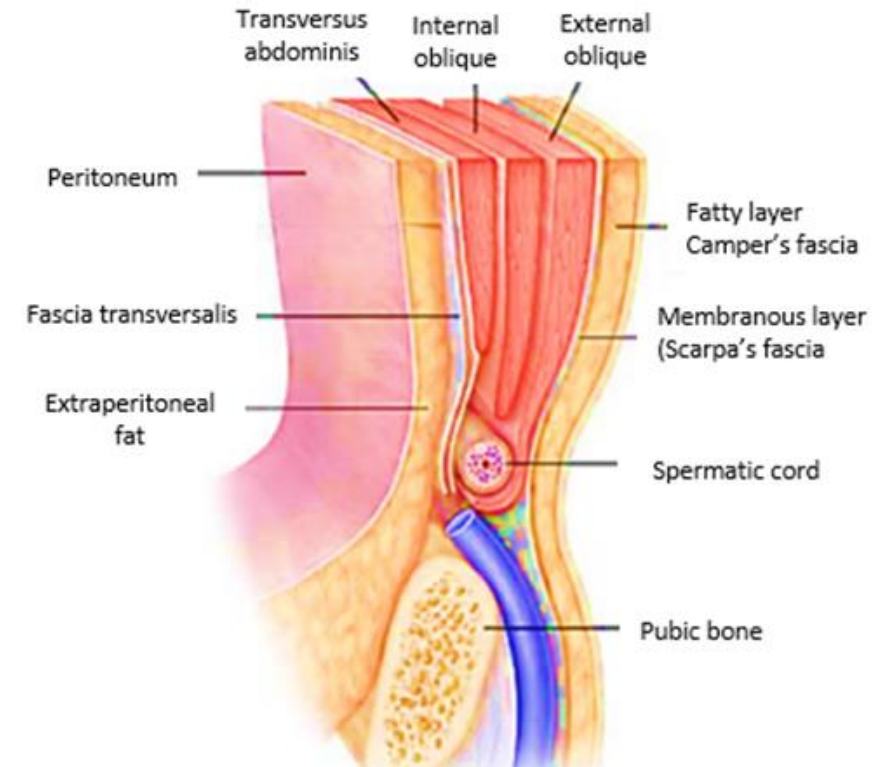
C. Below the level of the arcuate line:

- Anterior wall is formed by the aponeuroses of all three flat muscles.
- Posterior wall is deficient and is only formed by the fascia transversalis.



4. FASCIA TRANSVERSALIS

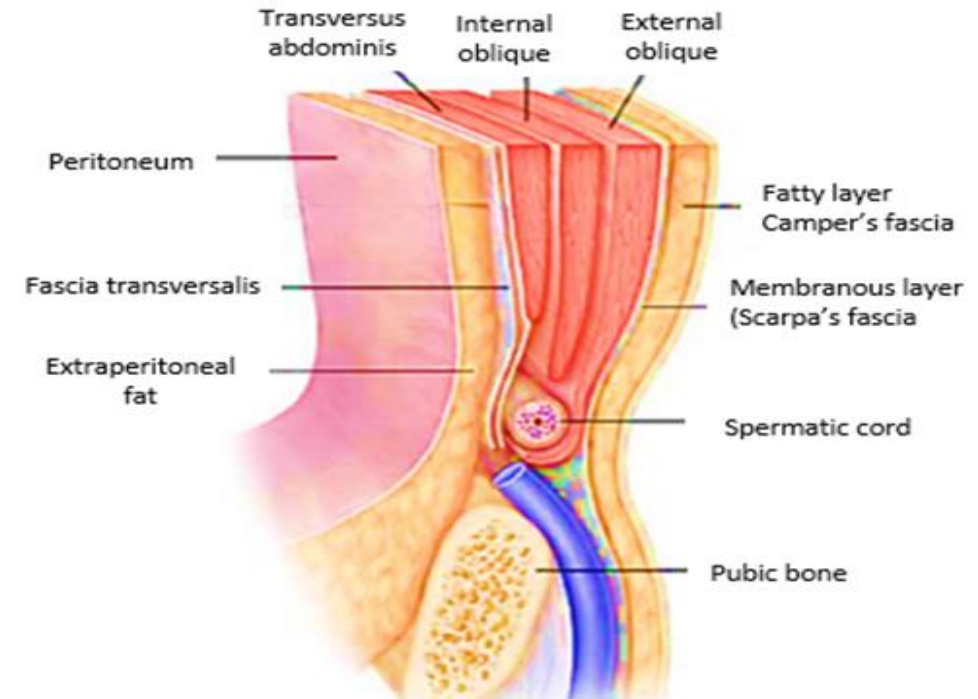
- ❑ The fascia transversalis is a thin fascia that lines the inner surface of the transversus abdominis muscle
- ❑ The fascia has an oval opening approximately 1.2 cm above the mid-inguinal point called the deep inguinal ring for the passage of the spermatic cord in males and the round ligament of the uterus in females.
- ❑ It forms a tubular prolongation around the spermatic cord known as the internal spermatic fascia.



5. EXTRAPERITONEAL FASCIA

Beneath the transversalis fascia lies a layer of connective tissue called the extraperitoneal fascia, which separates the transversalis fascia from the peritoneum

This layer, which contains varying amounts of fat, not only lines the abdominal cavity but also continues into a similar layer that lines the pelvic cavity.



6. PARIETAL PERITONEUM

Deep to the extraperitoneal fascia is the peritoneum, a thin serous membrane that lines the walls of the abdominal cavity at various points .

